

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15E.2

DETERMINING STRUCTURES USING MASS SPECTRA

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15E.2: Determining Structures Using Mass Spectra
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15E.2

Q1. 1. (a) The mass spectrum of an organic compound shows a molecular ion peak M^+ at $m/z = 58$ and a major fragment peak at $m/z = 43$. Identify the compound and the species responsible for $m/z = 43$.

[3 Marks]

WCH14/01/Oct21/Q8(a)

Q2. (b) Explain how the $M+1$ peak arises in mass spectra.

[2 Marks]

WCH14/01/Oct21/Q8(b)

SECTION B: WORKED MARK SCHEME — 15E.2

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) The mass spectrum of an organic compound shows a molecular ion peak M^+ at $m/z = 58$ and a major fragment peak at $m/z = 43$. Identify the compound and the species responsible for $m/z = 43$. 1. M^+ at $m/z = 58$ corresponds to propanal or propanone (C_3H_6O, $M_r = 58$). 2. $m/z = 43$ corresponds to $[CH_3CO]^+$ fragment ion (loss of $CH_3 = 15$). 3. Positive charge must be shown on fragment ion.	[3]
Q2	Q2: (b) Explain how the $M+1$ peak arises in mass spectra. 1. Caused by the naturally occurring carbon-13 (^{13}C) isotope (1.1% abundance). 2. Ratio of $M+1$ to M^+ peak height can be used to determine the number of carbon atoms.	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 15E.2

FAQ #1: Why must mass spectrum fragment ions carry a positive charge in equations?

Category: Mass Spectrometry

COMMON EXAMINER TRAP / MISTAKE:

Omitting the + sign on fragment formulas e.g. writing CH_3CO instead of $[\text{CH}_3\text{CO}]^+$.

MODEL ANSWER / EXAMINER EXPECTATION:

Only POSITIVELY CHARGED ions are accelerated and deflected by the magnetic field onto the detector. Uncharged free radicals are not detected.